

Qatar-4b

R-2267

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_241022 · created 2026-08-16T01:54:14+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460605.7939 +/- 0.0047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.03
Transit depth (Rp/Rs)^2	3.34 +/- 1.1 [%]
Orbital Inclination (inc)	84.43 +/- 1.92
Airmass coefficient 1 (a1)	0.913 +/- 0.019
Airmass coefficient 2 (a2)	0.073 +/- 0.056
Scatter in the residuals of the lightcurve fit is	2.13 %
Best Comparison Star	#1 - [221, 265]
Optimal Method	PSF photometry
Transit Duration (day)	0.063 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.183 ± 0.03 vs 0.138 (deviation 1.5σ)
Duration fit vs published	0.063 d vs 0.0892 d (deviation 1.87σ)
Mid-time O–C	0.4 min
Depth SNR	6.1
Residual scatter	2.13 %

Light curve

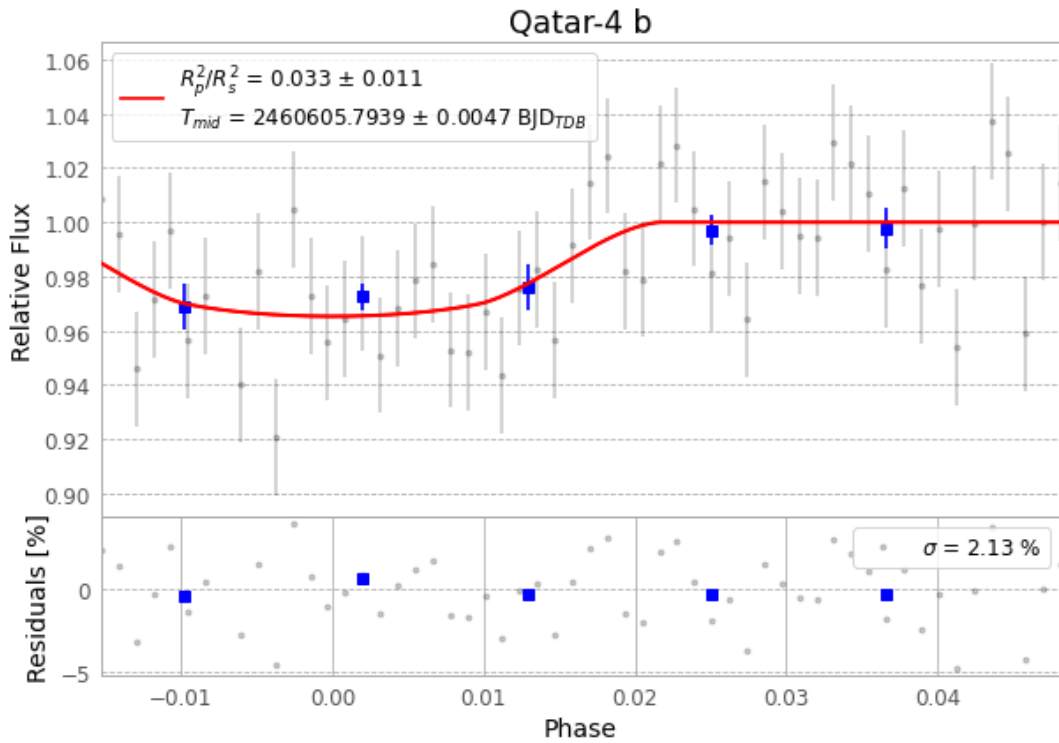


Figure 1 — FinalLightCurve_Qatar-4 b_2024-10-21.png (SHA-256 5926aabdb4efd21f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 276], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b7c9e86b46b38d6b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

57 FITS frames (37 MB), Qatar-4241022061216.FITS → Qatar-4241022090016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-241022.log (eef7608931ae...), FinalLightCurve_Qatar-4 b_2024-10-21.png (5926aabdb4ef...), FinalLightCurve_Qatar-4 b_2024-10-21.csv (3f02db41fa79...)

Compute resources

Wall time 102.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 01:54Z.